

# Thirumurugan Candane

## Full Stack Developer | Systems Engineer | Rust Developer | Mobile App Developer

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Full-Stack Developer with a strong focus on systems-level engineering, high-performance backend development, and scalable data pipelines. Experienced in building low-level solutions in Rust, including a custom read-only NTFS driver for efficient disk imaging and a multicast-based data transmission module for large-scale deployments.

Currently working at Zoho Corporation, contributing to real-time systems and high-throughput server architectures, with a focus on low-latency file transfer, concurrency, and resource optimization. Skilled in bridging system internals with modern application stacks, including Flutter-based mobile apps and RESTful backend services.

Strong interest in file systems, storage systems, and distributed architectures, with hands-on experience in handling large-scale data operations, compression, and efficient I/O workflows.

## WORK EXPERIENCE

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### Zoho Corp — Full Stack Developer

Oct 2022 – Present

- Engineered a **read-only NTFS driver in Rust** for disk imaging, eliminating constraints such as requiring identical source and target disk sizes, significantly improving flexibility and efficiency.
- Designed and implemented a **multicast-based data transmission module**, enabling efficient large-scale data broadcasting across multiple clients and improving deployment scalability.
- Architecting a **high-throughput upload server** optimized for large file transfers, focusing on **low latency, concurrency control, and efficient resource utilization**.
- Built and optimized **system-level data handling workflows**, improving I/O efficiency and memory utilization for large-scale operations.
- Developed and integrated **RESTful APIs** to support distributed system components and backend services.
- Collaborated across teams to deliver **performance-critical features**, enhancing system reliability, scalability, and maintainability.

### Enterprise Mind — Android Developer Intern

Jan 2022 - Jul 2022

- Developed cross-platform mobile applications using **Flutter**, integrating backend services via REST APIs.
- Implemented features leveraging **AI-driven logic**, improving application functionality and user experience.
- Collaborated with cross-functional teams to troubleshoot issues and deliver optimized solutions.

### GSK Technology — Python Developer

Jan 2021 - Feb 2021

- Built Python solutions for **real-time processing of CCTV footage and barcode scanner data**.
- Applied optimized **algorithms and data structures** to improve processing efficiency.
- Assisted in debugging, performance tuning, and maintaining technical documentation.

## SKILLS

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**Systems & Low-Level:** Rust, C++

**Backend:** Python (Flask, Django), Spring Boot, Node.js (Express), Java

**Frontend:** HTML, CSS, JavaScript, React, Ember.js, Svelte

**Mobile:** Flutter (Android, iOS)

**Databases:** PostgreSQL, Firebase

**Core Concepts:**

- File Systems (NTFS internals, disk imaging)
- High-throughput systems & concurrency
- REST API design
- Distributed systems basics
- Performance optimization

## PROJECTS

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**FinTrac:** Personal expense tracker built using Flutter and Supabase (PostgreSQL) with an offline-first architecture. Supports seamless expense tracking with local storage and secure data synchronization.

**AmSt (Real-Time Chat Application):** Developed a chat application using **Flutter and Firebase**, supporting real-time messaging with text, image, and video capabilities. Integrated Firebase Authentication and Cloud Firestore.

**LeafX:** Android application using **Flutter + Python Flask API** to predict plant diseases and suggest medicinal solutions based on input data.

## EDUCATION

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**Manakula Vinayagar Institute of Technology, Puducherry**

B.Tech in Computer Science and Engineering (2019 – 2023)

## ACHIEVEMENTS

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- Achieved **3rd and 4th level** in State-level International Informatics Olympiad (2017)
- Developed a **WhatsApp automation tool** using Python (Tkinter GUI)
- Built a **YouTube downloader** using Python (QML, PyQt5) with modern UI

## TARGET ROLES

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- Systems Engineer (File Systems)
- Backend Engineer (High-Performance Systems)
- Rust Developer
- Infrastructure / Platform Engineer